

Zernio × Tolley

The one social-publishing vendor under toley.io and toley.io/animate — what we use, what we skip, what it costs · August 19, 2026

Zernio (formerly Late / getlate.dev) is a **unified social-media API**: one key, 16 platforms, hosted OAuth, one `POST /posts`. We use exactly **two slices** of it — customer publishing inside **/animate** and Jared's **own channels** replacing three fragile browser-automation stacks. Everything else it sells (unified inbox, SMS/WhatsApp numbers, ads manager, comment-to-DM workflows) we deliberately do not use. This doc is the whole integration on four pages.

1

team API key
ZERNIO_API_KEY —
both lanes share it

2

lanes
/animate customers ·
Jared's channels

6+1

platforms routed
TT·IG·FB·Pin·X·LI +
YouTube native

18

owner accounts
connected & active
today

\$12

free credit / month
≈ 2 accounts before
billing starts

01 The mental model — four nouns, top down

TEAM
our key

One team = our `sk_...` API key

Billing pool, rate-limit budget and every profile below hang off this ONE key. Base URL `https://zernio.com/api/v1`, Authorization: Bearer . Same key in Vercel prod and `~/config/zernio/env`.

▼ contains

PROFILE
a folder

Profile = a named folder of accounts (24-char Mongo id)

/animate: one profile per customer, auto-created as `animate_<userId>`, id stored in `VaterSocialProfile`. **Owner lane:** Jared's hand-made profiles in the dashboard — Ruthann, Your KC Homes, Cordless, Businesss, Jelly Studio, WD Business, Estate Sales.

▼ contains

ACCT
a login

Account = one connected social login — the thing posts target

Born from a hosted-OAuth click. Mirrored into OUR db: `SocialAccount` rows (`provider='zernio'`, `externalAccountId`) for /animate; `~/config/zernio/accounts.json` for the owner lane. Health flags: `isActive`, `needsReconnection`.

▼ targeted by

POST
the work

Post = one `POST /posts` fanned out to N accountIds

Video by public URL (`mediaItems:[{type:'video',url}]`), per-platform targets + quirks, `publishNow` or `scheduledFor` + `timezone`, idempotent via `x-request-id`. Status flows back over ONE webhook.

The one security rule that is OURS, not Zernio's

Zernio validates an `accountId` against the **whole team**, not against a profile. If we ever pass customer B's `accountId` while acting for customer A, Zernio will happily post. The profile ↔ user and account ↔ user mapping lives in **our** database — every publish route must check `SocialAccount.userId` before calling the vendor.

02 Lane 1 — /animate customers connect their own socials

The product lane. Every /animate user connects their OWN TikTok / Instagram / Facebook / Pinterest / X / LinkedIn and publishes finished MP4s from the publish panel. Chosen 2026-08-17 over Ayrshare (\$599/mo for multi-user) and over native OAuth (5 separate platform app reviews).

1

Connect click → profile exists

ensureProfileForUser() — POST /profiles name animate_<userId> with Idempotency-Key ; on 409 it looks the profile up by exact name. Row upserted into VaterSocialProfile .

lib/vater/social-vendor/zernio.ts · profile names are unique per team

2

Hosted OAuth — Zernio does the consent screens

GET /connect/{platform}?profileId&redirect_url → authUrl . Zernio runs the platform consent plus any picker (FB page, IG account, Pinterest board, LinkedIn org), then 302s back to our callback. force=true re-runs OAuth for a reconnect. **Links embed a timestamp and EXPIRE — always generate fresh, never store or re-send one.**

app/api/vater/social-accounts/oauth/{platform}/start → callback

3

Callback → mirror accounts into our DB

syncAccountsForUser() — GET /accounts?profileId , upsert SocialAccount rows (status active / expired / error from needsReconnection / isActive) , delete vendor rows the vendor no longer reports. Never touches native YouTube rows. UI lands on /animate?social=connected#r=publishing .

4

Publish click → one createVideoPost()

POST /posts : caption, mediaItems:[{type: 'video', url: finalVideoUrl}], platforms:[{platform, accountId, platformSpecificData, customContent}], publishNow:true (or scheduledFor + IANA timezone), header x-request-id = stable id per (project, click) so a double-click can't double-post — the API returns existingPost instead.

app/api/vater/youtube/{id}/publish-social · ledger row: VaterSocialPost

5

Status comes back over the webhook, not polling

POST https://www.tolley.io/api/webhooks/zernio — HMAC-SHA256 of the raw body vs X-Zernio-Signature (refuses unsigned traffic), in-process dedupe on event id. Handles post.published / partial / failed / cancelled / scheduled / post.platform.* / post.tiktok.url_resolved / account.connected / account.disconnected → updates VaterSocialPost + re-syncs accounts. "Your posts" on the Publishing screen reads that ledger; GET /posts/{id} is the manual-refresh fallback.

Per-platform quirks we handle

X: customContent , first 270 chars.
Pinterest: needs boardId — boards via GET /accounts/{id}/pinterest-boards . **TikTok:** privacy_level + content_preview_confirmed / express_consent_given: true in top-level tiktokSettings ; options from .../tiktok-creator-info .

YouTube stays native

YouTube never touches Zernio — our own per-user Google OAuth (oauth/youtube/*). The publish panel fans one click out to native YT + Zernio platforms together.

Two hard rules

The MP4 must be a **public https Blob URL** — Zernio fetches it itself; DGX-tunnel links 409. And **nothing posts without the user's click** — no autonomous sends, ever.

Env (Vercel prod + .env.local)

ZERNIO_API_KEY (sk... team key) · ANIMATE_SOCIAL_VENDOR=zernio (feature flag — unset disables the whole lane) · ZERNIO_WEBHOOK_SECRET (HMAC). Migration 20260817_social_vendor ; webhook registration id 6a8390586ee5c447bf023243 .

03 Lane 2 — Jared's own channels (browser stacks → API)

Goal: retire the three fragile posting stacks — Selenium Pinterest (:9107/:9108), Selenium+Xvfb TikTok (:9106), Playwright LinkedIn — by routing those lanes through the same team key. Connects were completed 2026-08-19: **18 active accounts, zero needing reconnection.**

⚠ Where the accounts actually live (this bit surprised us)

Jared connected everything under profiles he made himself in the Zernio dashboard — **not** under the CLI's `owner_jared` profile, which sits empty. So `zernio_admin.py accounts` (profile-scoped) says "none connected"; the truth comes from team-wide `GET /accounts`. Automation must post with each account's **real** profileid from `accounts.json` — never assume one owner profile.

The 18 connected accounts, by profile

Businesss	TikTok @yourkchomes
Ruthann	Pinterest @digiegold · TikTok @ruthanntreasurehauls · FB RuthannsTreasureHaul · IG ruthanntolley · YT ruthanntreasures
Your KC Homes	Pinterest yourkchomes · FB page · YT yourkchomes · Meta Ads
Cordless	LinkedIn Jared Tolley · TikTok digitaljared · YT digitalgold
Jelly Studio	FB page · YT digitalgold-diggers · Meta Ads
WD Business	FB Wash & Dry Rental — Kansas City
Estate Sales	FB Tolley Estate Sales

LinkedIn: the Your-KC-Homes *page* posts ride the personal token's `w_organization_social` scope — one account row covers both.

The tooling (DGX-side, no site code)

- `~/growth-engine/lib/zernio_admin.py` — ensure-profile · connect <platform> (prints fresh hosted-OAuth URL) · accounts --write · boards <key> · post-status <id>.
- `~/growth-engine/lib/zernio_publisher.py` — post(account_key, caption=, media_url=|image_url=, board_id=, link=, title=, request_id=) . **DRY-RUN unless ZERNIO_PUBLISH=1**; every send appended to `~/growth-engine/state/zernio-posts.jsonl`.
- `~/config/zernio/accounts.json` (0600) — stable keys → real ids: `pinterest_main`, `pinterest_treasure`, `tiktok_kchomes`, `tiktok_ruthann`, `linkedin_jared` (+13 extras under `unmatched_extra`).
- `~/config/zernio/env` (0600) — same team key as toley-site.

Phase 3 — lane-by-lane cutover (unblocked, not started)

3a

Flag, don't rewrite

Each posting lane gets `PUBLISH_VIA=zernio|browser`; the browser path stays as fallback for 7 days per lane. Order: pin-featured → pin-circle → pin-haul/treasure-pins → retire the duplicate `pinterest-cron-post` cron → re-enable `LISTINGS_PINTEREST` → housing-hub LinkedIn → shorts TikTok leg → ruthann-repost (keeps its ramp/cap).

3b

Phase 4 — turn the browsers off

After 14 clean days: disable :9106/:9107/:9108 services + `tiktok-session-watchdog`. **Not moving:** FB / YT / IG / Bluesky legs, Marketplace, YT Community posts (no public API exists), tt-personal-picks (draft-gated).

04 What it costs — one pool for both lanes

Per connected account, graduated + prorated daily

\$6 first tier → **\$3** → **\$1** per account-month, applied to the TEAM total (billable units = all account-days ÷ days in month), not per individual account. Daily proration: a \$6 account connected 15 of 30 days costs \$3. **\$12 free credit each calendar month** ≈ two baseline accounts. Card auto-charges when accrued usage hits a threshold that starts at \$10 and doubles.

What this means for us

Owner accounts and every /animate customer's accounts **share one billing pool and one rate-limit budget** on the single team key. Zernio bills per connected account, **never per post** — so no per-post cost is booked onto Vater render cards. A 402 PAYMENT_REQUIRED anywhere = team-level billing alarm, not a customer problem.

Rate limits (scale with connected accounts)

0–2 accounts → 60 req/min · **3–2,000** → 600 req/min · **2,001+** → 1,200 req/min. We're in the 600 tier. Honor `Retry-After` on 429. Prefer the webhook over polling — polling burns this budget.

Webhook delivery contract

Respond **2xx within 5s** or Zernio retries on exponential backoff — 7 attempts (10s → 100s → ~17m → ~2.8h → 24h×2) then dead-letter. Delivery is **at-least-once**: dedupe on the event `id` / `X-Zernio-Event-Id`. `attemptNumber` tells you which retry you're seeing.

05 The gotcha list — every one already cost us time

- **Always `www.tolley.io` for the webhook** — the apex 301 redirect eats POST bodies.
- **Webhook registration endpoint is `/v1/webhooks/settings`**, not `/v1/webhooks`. Up to 10 endpoints per team.
- **Hosted-OAuth connect links expire** (timestamp baked in). Generate at click-time; a stored link in an /hq item or doc will be dead.
- **`accountId` is validated team-wide** — tenant isolation is our server-side check, page 1.
- **Media is fetch-by-URL** — the MP4/image must be publicly reachable https (Vercel Blob). Private or tunneled URLs fail the post.
- **Offboarding a customer:** `DELETE /accounts/{id}` per account, then `DELETE /profiles/{id}`. Disconnected accounts can be moved between profiles, never silently deleted.
- **Scoped API keys exist** (`POST /api-keys`, scope `profiles` + `profileIds`, read/write, optional expiry) — the right tool if we ever hand a tenant or a contractor direct API access. Nobody has one today.
- **Zernio's "AI providers" page is not for us** — it powers their comment → DM workflow bots only. Captions are generated DGX-side. If ever used: a fresh spend-capped Anthropic key, never the toley-site/OpenClaw keys.
- **Nothing posts autonomously** — /animate posts happen on the customer's click; owner-lane posts stay DRY-RUN until `ZERNIO_PUBLISH=1` and Jared has approved the lane.

Sources

docs.zernio.com (multi-tenant, webhooks, billing) fetched 2026-08-19 · `lib/vater/social-vendor/zernio.ts` · `app/api/webhooks/zernio/route.ts` · `~/growth-engine/lib/zernio_admin.py` / `zernio_publisher.py` · live team-wide `GET /accounts` 2026-08-19 (18 active) · memory: animate-zernio-social-publishing, zernio-owner-lane. Prices/limits are Zernio's published numbers on the fetch date — re-check the Pricing page before capacity planning.